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CS-2404

B. Tech. (CSE) (Third Semester)

EXAMINATION, 2025-26

COMPUTER ORGANIZATION

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any seven questions. Each question carries 8 marks. $7 \times 8 = 56$

- ✓ 1. Describe difference between 2D and 2½ D RAM ?
2. Explain Tristate buffer with example.

P. T. O.

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- ✓ 3. Describe difference between SRAM and DRAM.
4. Describe Register stack purpose in stack organisation.
- ✓ 5. Explain three Address Instruction formed.
- ✓ 6. Explain difference between RISC and CISC.
- ✓ 7. Explain Booth algorithm in brief.
8. Explain Displacement Addressing Mode.
9. Explain Flynn's classification.
- ✓ 10. Explain difference between multiplexer and decoder.

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 12 marks. $2 \times 12 = 24$

- ✓ 1. (a) Explain the difference between Direct Addressing Mode and Indirect Addressing Mode.
- (b) Explain Microoperation in brief.

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- ✓ 2. Explain 32 bit and 64 bit IEEE format convert (450.25) in 32 bit and 64 bit IEEE format.
3. Explain about Interrupt Driven in data transfer with their types and example.
4. Why BVS Arbitration required in organisation ? Describe any *one* type of Arbitration method.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

1. (a) Explain about direct mapping in cache.
 - (b) Consider a direct mapped cache of size 16 kB with block size 256 bytes. The size of main memory is 128 kB.
- Find :
- (i) Number of bits in tag.
 - (ii) Size of main memory in bits.

P. T. O.



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2. Explain about pipelining in computer organisation.
3. (a) Explain about I/O processor in computer organisation.
(b) Explain DMA controller.
4. (a) Explain hazard in computer organisation.
(b) Explain virtual memory.

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B. Tech. (Third Semester)

EXAMINATION, 2025-26

DIGITAL LOGIC DESIGN

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions. Each question carries 8 marks. $7 \times 8 = 56$

1. Design T Flip Flop using Logic Gates.

P. T. O.

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2. Explain the operation of half wave rectifier and derive the expression for its ripple factor and efficiency.
3. Design full adder and full subtractor by using Universal Gates.
4. Design on excess-3 to BCD code converter.
5. Explain the operation of a SR flip-flop.
6. Design a 4 bit combinational logic to subtract one bit from the other. Draw the logic diagram using NAND and NOR gates.
7. Construct a 16×1 multiplexer with two 8×1 and one 2×1 multiplexers. Use block diagrams.
8. Define RS Flip Flop.
9. Convert decimal 27.315 to binary and into hexadecimal.
10. Define encoder. List out the application of it.

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Section—B

(Short Answer Type Questions)

Note : Attempt any two questions. Each question carries 12 marks. $2 \times 12 = 24$

1. Design the four input multiplexers. Implement the following function $f(A, B, C) = \sum m(1, 3, 4, 6, 7)$ control variable B and C.
2. (a) Define multiplexer. Draw a $2 : 1$ multiplexer for the function $f(x, y, z) = \sum(0, 2, 3, 5, 7)$.
(b) Implement 4×16 decoder using two 3×8 decoders.
3. What is the Bipolar Junction Transistor ? Explain the different configurations of BJT.
4. What is the drawback of JK Flip-Flop ? Design a Flip-Flop which overcomes this drawback and explain with a neat diagram.

P. T. O.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

1. With the help of neat diagrams explain the operation and drain characteristics of N-channel MOSFET.
2. Reduce the following function using K-map.
$$F(A, B, C, D, E) = \sum m(1, 4, 8, 10, 11, 20, 22, 24, 25, 26) + d(0, 12, 16, 17).$$
3. (a) Explain a right shift register.
(b) Design a 3 bit ripple counter.
4. (a) Differentiate between combinational circuit and sequential circuit with a neat diagram.
(b) Differentiate between broadband and baseband transmission with advantages and disadvantages.

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BT-2465

B. Tech. (BT) (Third Semester)

EXAMINATION, 2025-26

DATA STRUCTURE AND ALGORITHMS

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any seven questions. Each question carries 8 marks. $7 \times 8 = 56$

1. What do you mean by term data structure ? Explain with suitable example.

P. T. O.

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2. What is Time complexity and Space complexity ?
3. What do you know about stack ? Explain.
4. Explain circular queue.
5. Write a program in 'C' to search a number in the array of 25 elements.
6. What is Infix, Prefix and Postfix in binary tree ?
7. What do you know about string ? Write name of any *three* functions of strings.
8. Write algorithm for POP operation.
9. What is extended binary tree ?
10. Give the proper definition of array and how memory get allocated to the array.

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 12 marks. $2 \times 12 = 24$

1. What do mean by bubble sort ? Explain with suitable example.

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2. Compare linked list and array.
3. Write a function insertion of a node in the last of linked list.
4. Write algorithm for selection sort. Also calculate its time complexity.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

1. What do you mean by quick sort ? Explain in detail.
2. What is Graph ? Explain about different types of graphs.
3. What is insertion sort ? Explain in detail.
4. What is Kruskal Algorithm ? Explain with suitable example.

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CS-2402

B. Tech. (Third Semester)

EXAMINATION, 2025-26

DATA STRUCTURE USING "C"

Time : Two Hours

Maximum Marks.: 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions. Each question carries 8 marks. $7 \times 8 = 56$

- ~~1.~~ Define Algorithm and write an algorithm to add two numbers.
- ~~2.~~ What do you mean by spanning tree ?
- ~~3.~~ Define term stack and its operation.

P. T. O.

Q9 What is Insertion Sort, Explain with the time complexity

Q10 What is Fibonacci Binary tree and complete Binary tree. [2] CS-2402

4. What is circular queue and how to write algorithm to form circular queue?
5. Write difference between contiguous and non-contiguous memory allocation with suitable examples.
6. Write a C-code to solve an given array by (Bubble sort).
7. Write a processor to add a item at middle of link list with suitable diagram with its algorithm.
8. What is Data Structure ? Also explain classification of data structure with explaining it.

Section—B

(Short Answer Type Questions)

Note : Attempt any two questions in maximum 150 words. Each question carries 12 marks. $2 \times 12 = 24$

1. Write an algorithm to implement stack of size N, using an array. The elements in the

stack are to be integers and its operation supported are push and pop with its condition to prove it.

2. What is graph ? Give all its terminology with proper example.
3. Write a C program for selection sort.
4. Write heap sort algorithm and sort the given array :

10, 20, 5, 18, 11, 8, 25, 12

Section—C

(Long Answer Type Questions)

Note : Attempt any one question in maximum 300 words. Each question carries 20 marks. $1 \times 20 = 20$

1. Write shorts notes on any two of the following :
 - (i) Underflow and overflow
 - (ii) Sequential file
 - (iii) Searching and sorting with example.

P. T. O.

2. 10 30 20 15 40 35 05

For above array sort it with Bubble sort and also find its worst, average and best case complexity with respect to comparison and swapping also.

3. What is circular queue ? Explain briefly and implement a queue with the help of doubly linked list.

4. Define Binary search tree, AVL tree, and B-tree. What are the differences among them. Explain your answer with suitable examples wherever required.

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CS-2403

B. Tech. (Third Semester)

EXAMINATION, 2025-26

**COMPUTER BASED NUMERICAL
METHODS AND STATISTICAL
TECHNIQUES**

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any seven questions. Each question carries 8 marks. $7 \times 8 = 56$

1. The solution of a problem is given as $x = 3.345$, and it is known that the relative

P. T. O.

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error is not more than 1%. Find to four decimal digits, the range of values within which the exact value of the solution must lie.

2. Compute the percentage error in the time period $T = 2\pi\sqrt{l/g}$ for $l = 1$ m if the error in the measurement of l is 0.01.

3. Find the real root of $2x - \log_{10} x - 7$ correct to three decimal places.

4. Find the real root of the equation $x = e^{-x}$ using the Newton-Raphson method correct to four decimal places.

5. Assuming that the following values of y belong to a polynomial of degree 4. Compute the next two values :

x	y
0	1
1	-1
2	1
3	-1

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4	1
5	?
6	?

6. Find the cubic polynomial which takes the following values :

x	$f(x)$
0	1
1	2
2	1
3	10

7. Use Gauss's forward formula to find a polynomial of degree four which takes the following values of the function $f(x)$:

x	$f(x)$
1	1
2	-1
3	1
4	-1
5	1

P. T. O.

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8. By the method of least square, find the straight line that best fits the following data :

x	y
1	14
2	27
3	40
4	55
5	68

9. Determine the constants a and b by the method of least squares such that $y = ae^{bx}$ fits the following data :

x	y
2	4.077
4	11.084
6	30.128
8	81.897
10	222.62

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10. Obtain a relation of the form $y = ab^x$ for the following data by the method of least squares :

x	y
2	8.3
3	15.4
4	33.1
5	65.2
6	126.4

Section—B

(Short Answer Type Questions)

Note : Attempt any two questions. Each question carries 12 marks. $2 \times 12 = 24$

1. From the following table of values of x and y , obtain dy/dx and d^2y/dx^2 for $x = 2.2$.

P. T. O.

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x	y
1.0	2.7183
1.2	3.3201
1.4	4.0552
1.6	4.9530
1.8	6.0496
2.0	7.3891
2.2	9.0250

2. Find $f'(4)$ from the following data :

x	$f(x)$
0	0
2	8
5	125
1	1

3. Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using :

(i) Simpson's 3/8 rule taking $h = \frac{1}{6}$

(ii) Weddle's rule taking $h = \frac{1}{6}$

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4. A river is 80 m wide. The depth ' y ' of the river at a distance ' x ' from one bank is given by the following table :

x	y
0	0
10	4
20	7
30	9
40	12
50	15
60	14
70	8
80	3

Find the approximate area of cross-section of the river using :

(i) Boole's rule

(ii) Simpson's 1/3 rule.

P. T. O.

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Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

1. From the Taylor's series, for $y(x)$, find $y(0.1)$ correct to four decimal places if $y(x)$ satisfies $\frac{dy}{dx} = x - y^2$ and $y(0) = 1$. Also find $y(0.2)$.
2. Solve the equation $\frac{dy}{dx} = x + y$ with initial condition $y(0) = 1$ by Runge-Kutta rule, from $x = 0$ to $x = 0.4$ with $h = 0.1$.
3. Solve by Picard's method, the differential equations :

$$\frac{dy}{dx} = z, \quad \frac{dz}{dx} = x^3(y + z)$$

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where $y = 1, z = \frac{1}{2}$ at $x = 0$. Obtain the values y and z from III approximation when $x = 0.2$ and $x = 0.5$.

4. Given $\frac{dy}{dx} = \frac{y-x}{y+x}$ with $y = 1$ for $x = 0$. Find y approximately by for $x = 0.1$ by Euler's method.

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